

PORTLAND TROUTDALE, OR, USA

WMO: 726985

Lat: 45.551N

Lon: 122.409W

Elev: 29

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 24242

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.4	29.4	10.8	9.6	29.5	16.8	12.9	32.7	32.6	37.5	30.9	38.0	17.6	90	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.1	91.6	67.4	88.1	66.5	83.9	65.1	69.5	86.9	67.9	84.6	66.2	81.4	6.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	87.5	73.3	61.8	82.6	72.5	60.6	79.0	71.8	33.5	87.4	32.1	84.4	30.8	81.5	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	25.3	22.7	DB	22.6	99.8	4.3	3.5	19.5	102.3	17.0	104.4	14.6	106.3	11.5	108.8	(4)
(5)			WB	20.6	72.2	4.5	1.7	17.4	73.5	14.8	74.5	12.3	75.5	9.1	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	41.2	43.6	47.7	52.0	58.2	63.4	69.3	69.5	64.0	55.1	47.1	40.9		
	DBStd	11.48	5.90	5.24	5.24	5.54	6.11	5.86	4.98	4.76	5.48	5.18	5.84	5.95		
	HDD50	1030	275	185	106	37	5	0	0	0	0	15	121	286		
	HDD65	4291	736	599	535	392	228	98	15	12	82	310	538	746		
	CDD50	2633	3	6	36	98	258	401	597	605	420	172	33	4		
	CDD65	421	0	0	0	3	16	49	146	153	52	2	0	0		
	CDH74	4777	0	0	0	39	270	643	1599	1565	642	19	0	0		
	CDH80	1848	0	0	0	5	77	250	660	646	209	1	0	0		
(14) Wind	WSAvg	7.0	12.6	10.3	7.6	5.8	4.5	4.2	3.9	3.7	4.1	5.9	9.5	11.7		
(15) Precipitation	PrecAvg	57.4	8.0	5.9	6.1	5.0	3.9	2.8	0.8	1.0	2.1	5.3	8.8	8.7		
	PrecMax	70.3	15.1	13.4	10.2	10.0	6.3	5.9	3.2	4.7	5.3	11.1	18.4	16.5		
	PrecMin	43.8	2.9	0.5	2.1	2.2	0.5	0.6	0.1	0.0	0.0	1.4	3.1	4.8		
	PrecStd	6.9	2.8	2.7	1.9	1.7	1.5	1.2	0.7	0.9	1.2	2.5	3.5	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	59.9	70.1	79.4	87.2	94.3	98.5	96.9	90.9	76.1	64.7	59.2		
		MCWB	53.0	51.4	56.5	60.3	64.3	68.4	70.1	68.5	65.6	59.6	58.1	54.9		
	2%	DB	54.4	56.1	64.0	72.7	81.6	87.5	91.4	91.4	85.8	71.9	61.1	54.5		
		MCWB	50.2	48.5	52.3	56.9	62.0	66.5	67.8	67.3	63.4	57.5	55.2	51.3		
	5%	DB	51.9	53.9	60.3	66.7	75.5	81.3	87.4	87.6	81.5	67.4	57.8	51.8		
		MCWB	48.1	47.5	50.3	53.5	59.7	63.7	67.0	66.8	63.1	56.3	53.0	48.6		
	10%	DB	49.1	51.6	56.7	62.9	70.8	75.9	82.4	82.4	76.9	63.9	55.1	48.3		
		MCWB	45.8	46.1	49.1	52.2	57.7	61.3	65.3	65.3	61.3	55.3	51.1	45.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	53.4	57.6	61.6	66.6	69.9	73.1	71.8	68.7	63.9	60.1	55.6		
		MCDB	57.7	57.0	68.5	77.1	84.1	89.6	92.6	89.6	85.7	70.9	62.9	57.9		
	2%	WB	50.9	50.6	54.0	57.8	63.2	67.3	69.7	69.4	66.2	60.4	56.1	51.6		
		MCDB	53.7	54.4	60.7	70.1	78.2	85.0	86.5	86.6	80.5	66.7	59.6	54.2		
	5%	WB	48.3	48.4	51.9	55.2	61.0	64.5	67.9	67.7	64.3	58.7	53.9	48.5		
		MCDB	51.0	52.4	57.9	64.5	73.4	78.9	84.5	83.9	77.0	64.2	57.1	51.1		
	10%	WB	46.1	46.5	50.2	53.0	58.8	62.4	66.1	66.0	62.5	56.9	52.0	46.0		
		MCDB	48.5	50.1	56.0	60.4	68.9	74.3	81.4	80.6	73.7	62.4	55.2	48.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	11.4	14.7	17.0	18.9	19.9	23.8	24.1	22.3	15.7	10.1	7.9		
		MCDBR	12.5	14.9	21.6	26.9	29.9	31.9	33.2	33.5	31.3	22.7	12.5	10.9		
		MCWBR	9.4	9.3	12.3	14.2	14.1	13.5	12.8	12.7	13.2	11.7	8.0	8.7		
	5% WB	MCDBR	12.0	12.8	18.0	22.7	25.9	28.7	29.9	29.7	27.3	17.2	11.1	10.7		
		MCWBR	9.6	8.6	11.1	12.7	12.8	12.6	12.4	12.0	12.3	10.2	8.0	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.316	0.329	0.365	0.378	0.361	0.351	0.358	0.345	0.335	0.319	0.325		
	taud		2.525	2.549	2.526	2.389	2.375	2.438	2.487	2.488	2.502	2.539	2.546	2.474		
	Ebn at Noon		252	277	288	284	282	287	289	283	278	265	250	234		
	Edh at Noon		21	24	29	36	37	35	33	32	29	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		370	641	939	1369	1737	1926	2182	1861	1390	807	436	302		
	RadStd		59	90	119	125	163	171	146	122	117	98	58	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+47

Nomenclature: See separate page